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Unlocking the Power of Social Capital: Advanced Strategies for Enhanced Personalized Recommendations in Online Social Networks

Completed Research Full Paper

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Abstract

In today's digital landscape, online social networks have become pivotal platforms for communication, content sharing, and information seeking. However, the huge amount of content has led to information overload, posing a significant challenge for users. This paper introduces a novel recommender system designed to mitigate this problem by integrating the concept of social capital into the recommendation process. Social capital, encompassing trust, influence, and relationships within social networks, is leveraged to enhance the relevance and personalization of news recommendations. By analyzing news with high user interactions and influence propagation, the system ensures that users receive content aligned with their preferences. The proposed methodology is detailed, and experimental evaluation demonstrates its effectiveness.

Keywords

Social Capital, Recommender Systems, Online Social Networks

Introduction

In recent years, Online Social Networks (OSNs) provided a wide environment for communication, sharing content, creating connections to socialize with users, or even searching information. Although these benefits of communication and accessing information are particularly evident in their role in spreading news and influence, we can notice the growth of the published content over the long years, overloading users most of the time. For instance, on the microblogging X also known as Twitter, over 500 million tweets are sent daily by its 229 million active users, creating a huge amount of information that is consumed by millions of other users.

Generally, people create online relations for many reasons such as to follow friends or family members, coworkers, or anyone who creates interesting content according to their preferences. However, they want to consume relevant content, not everything they post. In other words, they do not want to unfollow them, they want to receive personalized content considering their preferences. These observations tackle a significant issue prevalent in online social networks: Information Overload (IO). This refers to the inherent inability of humans to handle the vast amount of information generated on OSNs. Thus, we can notice that it is necessary building intelligent systems to filter information for suggesting relevant content according to the individual's preferences (Wu et al. 2022).

Recommender Systems (RSs) have emerged as a critical tool in managing the vast content present in OSNs, offering a solution to the challenge of filtering and presenting relevant news content. These systems employ techniques to analyze user behaviors, preferences, and interactions, thereby enabling the customization of

news feeds. This customization not only elevates the user experience but also addresses the generalized issue of IO in OSNs. By selectively presenting relevant news, RSs ensure that users engage primarily with content that aligns with their interests and needs. Many methods are used such as content-based filtering, which uses item features to recommend similar items based on a user's past behavior or explicit feedback, and collaborative filtering, which bases its suggestions on the interactions and data gathered from other users (Ricci et al. 2015). However, these traditional techniques create a key issue where users may be overwhelmed by the volume of content that is relevant but not necessarily aligned with their broader interests, leading to a paradox of choice. This can result in users facing an overload of content, making it challenging to select what is most meaningful to them (Lunardi et al. 2020).

An approach for improving the quality of recommendations in OSNs is the exploitation of Social Capital (SC), which involves leveraging the trust, influence, interactions, and relationships within a user's social network. By integrating these social dynamics, RSs can offer more nuanced and contextually appropriate content, thus improving the overall user experience. This approach acknowledges the multifaceted nature of relevance, going beyond algorithmic predictions to incorporate the social context in which these recommendations are made (Singh et al. 2022). The Social Capital definition has been conceptualized by Bourdieu (2002) as the cumulative worth of resources within a network of connections. The notion of SC is predicated on the idea that individuals gain from the various norms and values fostered and generated by a social network, such as trust, reciprocity, information, and cooperation, as noted by Spratt et al. (2019) and Coleman (1990). The gathering of SC via interactions and relationships within a social network can yield insightful data regarding a user's interests and preferences.

This paper delves into variables capable of improving recommendation quality, focusing on understanding, modeling, and leveraging the dynamics and resources present within social networks by utilizing social capital. The main goal is analyzing news that garner significant interactions, comments, and are propagated by influential users within OSNs. This methodology is designed to identify and recommend the most pertinent news to users, ensuring they are aligned closely with their interests.

This paper is structured as follows: Section 2 discusses related work. Section 3 depicts the proposed recommender system. Section 4 presents the experimental evaluation. Finally, Section 5 concludes this work and presents future works.

Related Work

In the realm of online social networks, the quest for personalized content recommendation has driven numerous studies, employing methods like Collaborative Filtering (CF) and Machine Learning (ML) to suggest relevant content based on user relationships and published content. However, a significant gap exists in leveraging Social Capital to enhance the quality of recommendations, indicating that there are a lot of opportunities for researching and applications.

Recent contributions to the field, such as Chang et al. (2022), propose a hybrid CF and social network analysis method for travel recommendations, yet only lightly explore SC techniques. Unlike these approaches, our study delves deeply into SC, integrating it with Content-based Filtering (CB) and social network analysis to harness the full potential of social interactions for relevant recommendations. Similarly, Singh et al. (2022) introduce a machine learning approach for improving Twitter friend suggestions using network analyses but only marginally apply SC techniques. Our research distinguishes itself by making SC a base of our system, enriching recommendations with profound social insights.

Further, studies like Logan et al. (2023) offer sophisticated analyses of Twitter interactions through directed multilayer network approaches but only skim the surface of integrating SC. Our work, in contrast, employs SC to understand the impact of trust and social connections on recommendations. Subramani et al. (2023) explore influencer prediction using a gradient boosted decision tree approach without deeply integrating SC concepts, whereas our study incorporates SC to reveal how influencers affect OSNs.

Chakrabarti et al. (2023) and Zhang et al. (2023) focus on hashtag recommendation and the dynamics of online news media, respectively, providing valuable tools and insights yet minimally engaging with SC. Our

research advances these discussions by examining how social dynamics and trust influence content engagement and information dissemination within OSNs. Additionally, Sun et al. (2023) and Li et al. (2023) introduce innovative approaches to group recommendation and user influence ranking, contributing to the understanding of social influence and group dynamics but lightly touching upon SC's explicit integration.

Our study stands out by deeply embedding SC into our analysis, offering a nuanced perspective on recommendations. By combining SC analysis with content-based filtering and social network analysis, we aim to develop a recommendation system that is not only contextually aware and personalized but also capitalizes on the social relations and interactions within OSNs. This approach seeks to unlock the untapped potential of SC, providing a richer, more targeted recommendation experience that aligns with users' preferences and social contexts, thereby addressing the limitations of existing methodologies and offering a novel pathway to enhance OSN user engagement.

The Proposal: The Social Capital Recommender System

This section describes the RS proposed by this paper where it is an extension of our last proposal De Souza and Durão (2024). It begins with an overview of the recommendation system. Subsequently, it introduces the key features of Twitter. The next part models news and users, leading into an examination of user influence and reputation. The final part outlines the algorithms of the recommendation model.

Recommendation Overview

The proposed recommendation system leverages Twitter data to curate suggestions, employing multiple metrics to evaluate the social significance of Twitter users and their posts. This includes assessing users' influence, reputation, recency, and context scores, which are derived from factors such as follower counts, tweet engagement rates, sentiment analysis of mentions and replies, as well as the relevance and diversity of tweet content, including the inclusion of pertinent media like images or videos links. These evaluations contribute to a comprehensive SC score for each tweet, enabling the system to prioritize and recommend the most relevant tweets to users. A visual representation of this RS is depicted in Figure 1.

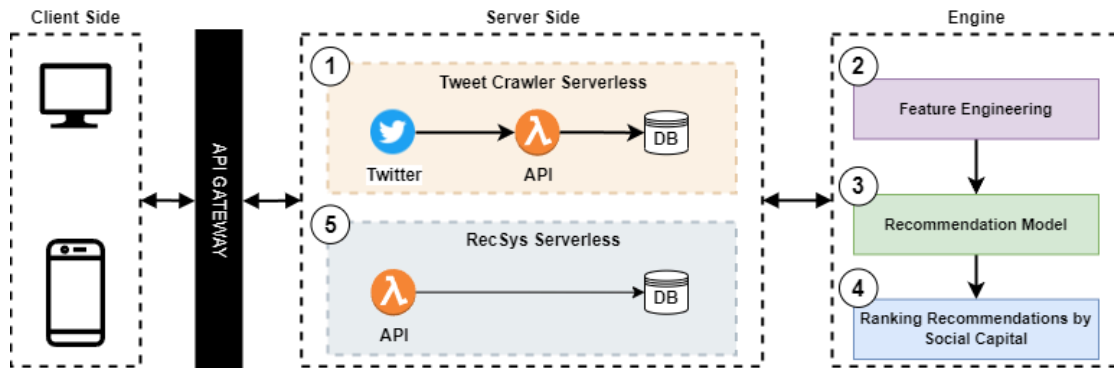


Figure 1. Recommendation Overview, which shows the macro steps to illustrate how the social capital recommendations are generated.

Firstly, tweets are extracted from the Twitter API and stored in the database to be used as input for the all-encompassing process, which is divided into three major steps: Feature Engineering, Recommendation Model, and Ranking Recommendations using the Social Capital Model. Finally, the generated recommendations are presented to the active user in descending order.

Feature Engineering

These steps play a vital role in cleaning and normalizing the unprocessed text, diminishing its complexity, and isolating the pertinent details necessary for modeling tweets. An overview of the methodologies employed in this phase is depicted in Figure 2.

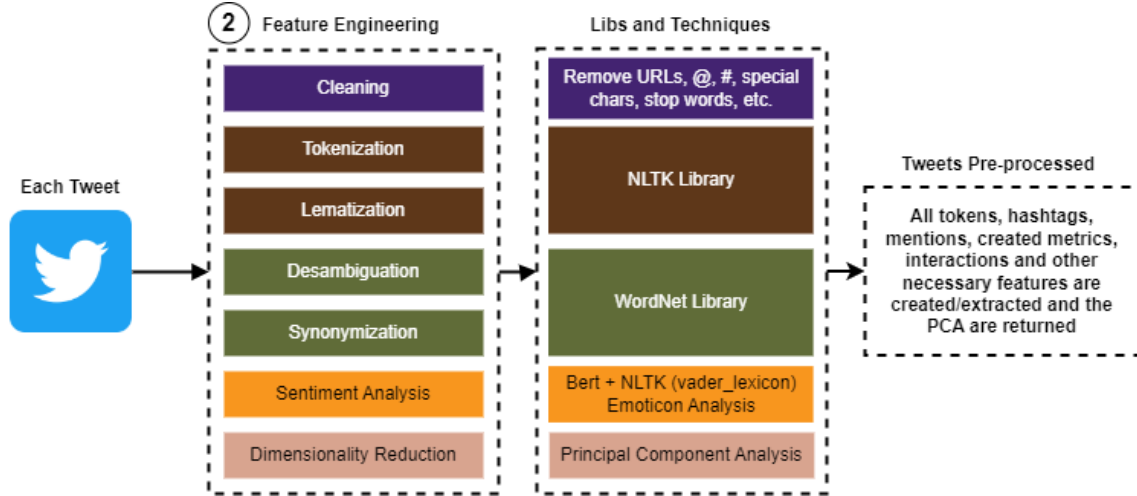


Figure 2. Feature engineering to process each tweet's text and relevant metadata.

Feature engineering in text analysis begins with converting all text to lowercase for standardize and removing URLs, mentions, and punctuation to focus on relevant content. This cleaning process pares down the text to its essentials for clearer analysis. Tokenization then breaks the text into individual words or tokens for detailed examination. By removing stop words, the process eliminates common words that contribute little to the overall meaning, emphasizing significant words. Lemmatization simplifies words to their base form, reducing word diversity and aiding in the grouping of related terms. Disambiguation addresses words with multiple meanings, ensuring accurate interpretation based on context. Including synonyms enriches the analysis by capturing the full essence of the text and grouping related concepts. To manage the high dimensionality of the resulting $m \times n$ matrix, where m represents tweets and n represents features, Principal Component Analysis (PCA) is employed to reduce dimensionality while preserving data variability, streamlining the feature space for more efficient analysis.

User's Metrics

To measure how influent and powerful a user can be within the network, we applied the **userStrengthScore** proposed by De Souza and Durão (2024). In summary this metric quantifies a social media user's influence by combining their reputation, based on positive and negative interactions, with their reach and engagement levels, factoring in follower count and activity engagement to assess their overall impact and authority on the platform.

Tweet's Metrics

In this work we applied the Diversity and Context metrics proposed by De Souza and Durão (2024), adopting a new Recency Score (Equation 1) for employing an exponential decay model aiming to moderate the rate of decay over time. The exponential approach, by using λ , provides a more intuitive control over how quickly the recency score decreases, offering a straightforward method to model the diminishing impact of content over time, potentially leading to a more nuanced and responsive recency scoring in dynamic datasets. This positive adjustment allows for a more refined tuning of the recency's influence on content relevance, enhancing the ability to prioritize newer content effectively in real-time analyses.

$$\text{RecencyScore}(t) = e^{-\lambda \cdot \text{elapsed}_{\text{OBJ}}} \quad (1)$$

where e is the base of the natural logarithm, λ represents the rate of decay over time (in this context, $\lambda = 0.03$). We choose this value empirically varying between 0-1. elapsed is the time elapsed in days between the most recent content's creation date in the dataset and the creation date of each tweet (t).

The Social Capital Metric

Equation 2 adopts a comprehensive methodology to assess a tweet's impact by integrating sentiment analysis (SI) calculating the sentiment impact, user engagement metrics (ES) as the average of likes, retweets, replies, and quotes, content relevance (CR) using TF-IDF scores, network connectivity (NI) based on mentions and URLs, Author's Influence (AI) as Count(Mentions), and Content Virality (CV) as Count(Retweets) + Count(Quotes). Here, AI is defined by the count of mentions, whereas CV encompasses the combined total of retweets and quotes. This equation delves into the emotional resonance of content through sentiment analysis, user engagement through likes and retweets, the distinctiveness of content via TF-IDF scores, and the breadth of content dissemination and author's network reach through CV and AI, respectively. Such an integrated approach ensures a nuanced evaluation of a tweet's potentials way across social networks by leveraging varied data dimensions. Take into consideration that the SI, ES, CR, and NI are discussed into this work (De Souza and Durão 2024).

$$SC = \alpha.SI + \beta.ES + \gamma.CR + \sigma.NI + \epsilon.AI + \zeta.CV \quad (2)$$

where $\alpha, \beta, \gamma, \sigma, \epsilon, \zeta$, represent the weights allocated to each component, cumulatively equating to 1. These weights are modifiable, promoting customized analyses to highlight specific facets of SC deemed pivotal. Adopting a value of 0.2 for each weight aims for an equitable contribution from each dimension, assuming an equal impact on a tweet's influence and network visibility. This choice is grounded in the desire for initial neutrality, simplicity in the model's interpretation, and the adaptability for weight adjustments based on analytical findings or specific research objectives. This balanced approach depicts the social media influence, emphasizing the importance content quality and the user's network position.

Recommendation Model

The recommendation model capitalizes on tweets' social capital scores to curate personalized content for users on OSNs, premised on the notion that tweets with higher engagement and visibility align more closely with user interests. The process initiates with a collection of tweets, followed by feature engineering process to refine the data. Subsequently, the SC of each tweet is assessed, forming the basis for generating a tailored list of recommendations, thereby enhancing the relevance of the content presented to the user.

The Social Capital Algorithm

The Algorithm 1 calculates the SC of a news tweet on a social media platform. The algorithm takes in a news *tweet* as input and returns its social capital score.

Algorithm 1: Enhanced Social Capital Score Calculation

Input: tweet

Output: social_capital_score

1. Normalize 'sentiment_score' to a scale of 0-1
2. Estimate Network Influence (NI)
3. Count non-null 'mentions' and store in 'mentions_count'
4. Count non-null 'urls' and store in 'urls_count'
5. Sum 'mentions_count' and 'urls_count' to get 'network_influence'
6. Calculate Content Virality (CV)
7. Sum 'retweet_count' and 'quote_count' to get 'content_virality'
8. Adjust Social Capital Score (SCS)
9. Define weights for SI, ES, CR, NI, AI, CV as 0.20, 0.20, 0.20, 0.20, 0.10, 0.10 respectively
10. Compute social_capital_score using the defined weights and component scores

11. **return** social_capital_score

Initially, on line 1, it normalizes the 'sentiment_score' between 0 and 1 to ensure uniformity across data points. Following this, the estimation of Network Influence begins, with lines 3 and 4 quantifying 'mentions' and 'urls' present in the content, which are then aggregated on line 5 to derive the 'network_influence' metric. The process of calculating Content Virality is initiated on line 6, with 'retweet_count' and 'quote_count' combined on line 7 to assess the content's spread across the network. An adjustment to the Social Capital Score (SCS) is prepared on line 8, where specific weights are defined on line 9 for Sentiment Impact, Engagement Score, Content Relevance, Network Influence, Author's Influence, and Content Virality, highlighting a balanced contribution from each factor. The 'social_capital_score' is computed on line 10 using these defined weights and component scores, culminating in the output of the 'social_capital_score' on line 11, offering a quantitative measure of the content's potential influence and relevance within the social network.

Experimental Evaluation

We present an offline evaluation to compare our proposal with several literature methods.

Dataset

This dataset was built with the assistance of 80 volunteers through an online experiment previously in this work (Souza et al. 2022) and then, this dataset was used in this other work (De Souza and Durão 2024). In addition, it encompasses a wide range of social media interactions and elements, specifically within the context of Twitter. A total of 3,080 recommendations were analyzed, derived from the activity of 12,185 Twitter accounts. It provides an in-depth look into user behavior on Twitter, including interactions with 9,533 mentioned users, which is consistent across several metrics: replies, retweets, URLs shared, hashtags used, and quotations, all involving 9,533 users each. Additionally, a huge number of tweets, totaling 32,252, were extracted (<https://github.com/pauloprdsouza/social-capital>).

Methodology

The offline experiment utilized a specific dataset to evaluate a newly proposed algorithm by running it across various combinations of user IDs and algorithms. The objective was to generate and then compare a new set of recommendation rankings against previously evaluated ones. This procedure was structured into three main steps: executing the algorithm to produce new recommendations for each user, comparing these new rankings with prior evaluations to determine the quality of the recommendations, and finally analyzing the results to ascertain the effectiveness of the proposed algorithm in enhancing the recommendation rankings. This methodical approach facilitated a detailed assessment of the algorithm's impact on the recommendation system.

Comparison Methods

The enhanced methodology underwent a comparative analysis against benchmarks established in our preceding study (see De Souza and Durão, 2024) and was juxtaposed with the existing baseline, which is recognized as the state-of-the-art in this work. **SCSA_PLUS**: Our last offline experiment; **STATE_ART**: Baseline (Tiwari et al. 2021); **SCSA_PLUS_V3**: Our proposal RecSys.

Ranking Metrics

To evaluate the recommendation algorithm, key metrics including Mean Reciprocal Rank (MRR) (Abel et al. 2013), Mean Average Precision at N (MAP@N) (Pla Karidi et al. 2018), Precision at N (Precision@N) (Abel et al. 2013) and Normalized Discounted Cumulative Gain at N (NDCG@N) (Zhang et al. 2015) were utilized. MRR focuses on the placement of the first relevant suggestion, highlighting the value of top rankings. MAP@N gauges the precision and relevance of the top N recommendations, while Precision@N measures the proportion of relevant suggestions within the top N items, reflecting the algorithm's precision.

NDCG@N evaluates ranking quality by rewarding highly relevant items positioned higher on the list, offering a comprehensive analysis of the algorithm's performance in delivering relevant and well-ranked recommendations.

Results

The results reveal nuanced insights into the performance of various recommendation models, as illustrated across Figures 3 to 10.

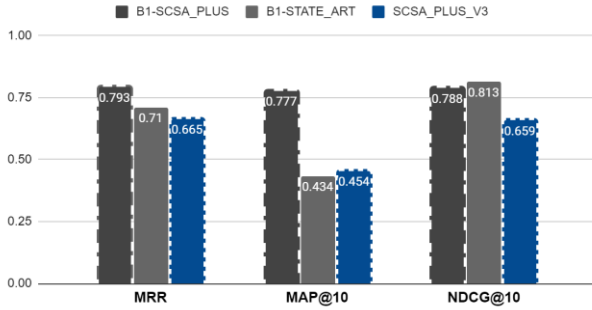


Figure 3. Baselines vs Proposal.

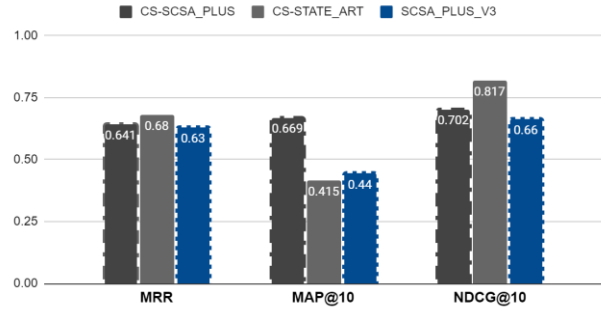


Figure 4. Baselines vs Proposal.

Figure 3 shows that the B1-SCSA_PLUS model outperforms others in MRR and MAP@10, indicating superior first relevant recommendation placement and precision within the top 10. However, B1-STATE_ART leads in NDCG@10, suggesting it ranks highly relevant items more effectively despite lower overall precision and recommendation placement compared to B1-SCSA_PLUS. SCSA_PLUS_V3 trails in all metrics, highlighting areas for improvement. In addition, Figure 4 depicts that CS-STATE_ART excels in NDCG@10, indicating it effectively ranks highly relevant items, despite lower MAP@10. CS-SCSA_PLUS shows balanced performance across all metrics, while SCSA_PLUS_V3 needs improvement in ranking and precision of top recommendations.

Figure 5 presents that SC-SCSA_PLUS demonstrates strong overall performance, leading in MRR and MAP@10, indicating high initial relevance and precision within the top 10 recommendations. SC-STATE_ART, despite a lower MRR and MAP@10, excels in NDCG@10, suggesting it ranks highly relevant items effectively at top positions. SCSA_PLUS_V3 trails in all metrics, indicating areas for potential enhancement. After that, Figure 6 shows that SCSA-SCSA_PLUS presents a balanced and strong performance across MRR, MAP@10, and NDCG@10, indicating effective overall recommendation relevance and ranking. SCSA-STATE_ART, while lagging in MRR and MAP@10, demonstrates a superior ability to rank highly relevant items with its higher NDCG@10 score. SCSA_PLUS_V3 exhibits lower performance in all metrics, suggesting areas for improvement.

Figure 7 shows B1-SCSA_PLUS consistently outperforms the other models across all precision metrics (P@1 through P@5), indicating it reliably identifies relevant recommendations at the top of the list. B1-STATE_ART shows improvement as more recommendations are considered, but still lags behind B1-SCSA_PLUS. SCSA_PLUS_V3 starts strong at P@1 and P@2 but sees a decrease in precision as more recommendations are evaluated, indicating variability in its ability to maintain relevance across the top recommendations. In turn, Figure 8 presents that CS-STATE_ART and SCSA_PLUS_V3 show strong initial precision at P@1, outperforming CS-SCSA_PLUS. However, CS-SCSA_PLUS demonstrates steady improvement from P@1 through P@5, indicating increasing relevance with more recommendations. CS-STATE_ART maintains consistent performance across all levels, suggesting a balanced recommendation quality. SCSA_PLUS_V3 has a notable drop at P@3, suggesting an inconsistency in maintaining relevance, but recovers at P@4 and P@5, closely aligning with CS-STATE_ART's performance.

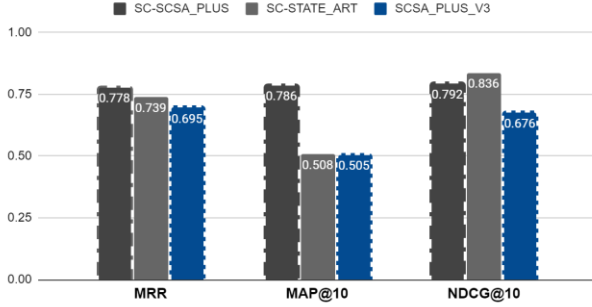


Figure 5. Baselines vs Proposal.

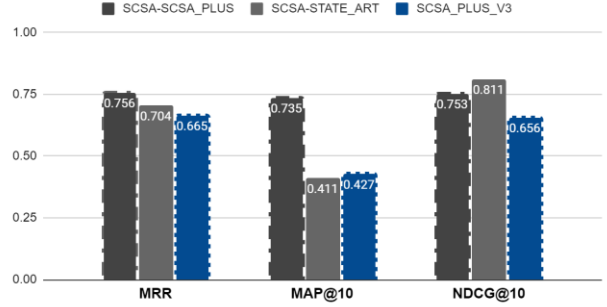


Figure 6. Baselines vs Proposal.

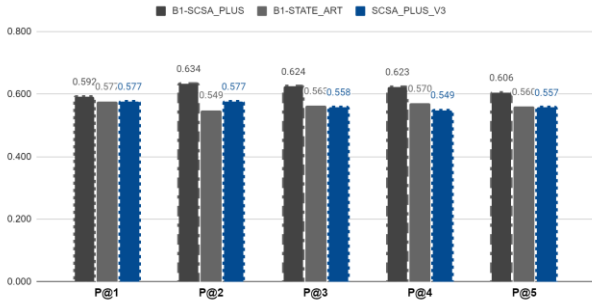


Figure 7. Baselines vs Proposal.

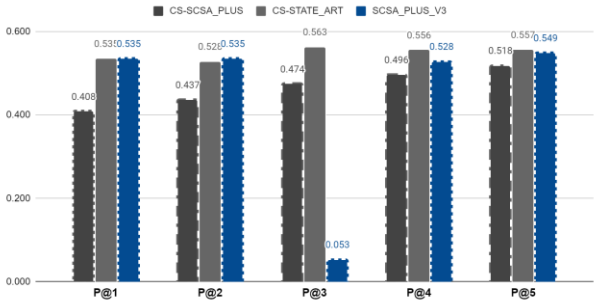


Figure 8. Baselines vs Proposal.

Figure 9 demonstrates that SC-SCSA_PLUS exhibits strong and consistent performance across all precision levels, peaking at P@5, indicating its effectiveness in identifying relevant recommendations throughout. SC-STATE_ART starts strong but sees a slight decline from P@3 to P@5, suggesting a minor drop in relevance with more recommendations. SCSA_PLUS_V3 shows remarkable consistency, with a slight fluctuation in precision, indicating a steady ability to maintain relevance across the ranked list, closely competing with SC-STATE_ART in overall precision. Finally, Figure 10 shows that SCSA-SCSA_PLUS demonstrates a gradual decline in precision from P@1 to P@4, with a slight improvement at P@5, indicating a decrease in the relevance of recommendations as the list extends. SCSA-STATE_ART starts with consistent precision at P@1 and P@2 but experiences a more pronounced drop from P@3 onwards, suggesting a significant decrease in relevance beyond the top two recommendations. SCSA_PLUS_V3 shows a drop in precision from P@1 to P@3, stabilizes slightly at P@4, and then dips again at P@5, indicating inconsistency in maintaining relevance across the recommendation list. Overall, all models start similarly at P@1 but diverge in their ability to maintain precision with an increasing number of recommendations.

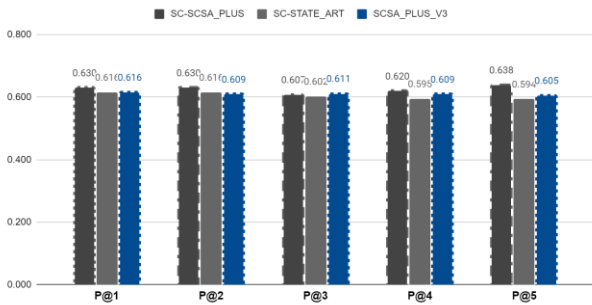


Figure 9. Baselines vs Proposal.

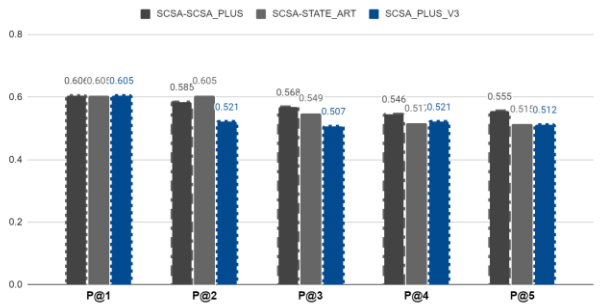


Figure 10. Baselines vs Proposal.

Discussion

This work explores the intricate dynamics of OSNs, and the pervasive challenge of information overload faced by users amidst the vast sea of digital content. Central to our investigation is the deployment of RSs

as a strategic solution to navigate and personalize the overwhelming content landscape. By leveraging advanced algorithms that analyze user behaviors, preferences, and interactions, RSs have demonstrated significant potential in tailoring news feeds to individual users. This customization not only enhances the user experience by presenting content of genuine interest but also addresses the broader issue of IO by filtering out irrelevant information.

Among the various techniques employed, CB filtering and CF stand out for their ability to recommend items like those a user has previously engaged with or to suggest content favored by similar users. These methods, rooted in algorithmic predictions, are instrumental in identifying relevant content. However, they sometimes fall short by presenting an abundance of choices, potentially leading to a paradox of choice. To mitigate this, our approach integrates Social Capital (SC) into RSs, emphasizing the value of trust, influence, and network relationships. This novel strategy enriches recommendations with a layer of social context, acknowledging that relevance transcends mere algorithmic predictions.

Limitations

The study delves into recommender systems, noting its limitations and proposing avenues for improvement. It focuses solely on offline evaluations, overlooking real-time dynamics of social media. Incorporating online evaluation methods could offer insights like click-through rates and dwell time. The model lacks knowledge-enhancement techniques, limiting its contextual understanding. Integration with external databases could enrich user profiles, enhancing recommendation relevance. Neglecting user community dynamics hinders capturing community influence on preferences. Algorithms detecting and utilizing community dynamics could boost recommendation accuracy. Future research should prioritize user-centric metrics and address these limitations to refine recommender systems. This would contribute valuable insights, especially in adapting to the diverse and dynamic landscape of social media platforms.

Conclusion

In conclusion, this study underscores the pivotal role of Social Capital in refining recommendation systems within OSNs. By integrating SC with established techniques like content-based filtering, we propose a novel framework that not only personalizes content suggestions but also deeply respects the intricacies of user interactions and social dynamics. Our approach marks a significant departure from traditional methods, promising to enhance user experience by delivering more relevant and socially contextualized content. This methodological innovation opens new avenues for leveraging social connections and trust in shaping the digital content landscape.

Looking ahead, future work will focus on further refining the SC-based recommendation framework, exploring advanced analytics to better understand and quantify the nuances of social interactions. We aim to incorporate real-time data analysis for dynamic recommendation adjustments, adapting to the evolving nature of user preferences and social ties. Additionally, we plan to extend our research to encompass a wider range of OSNs, investigating the applicability and impact of our SC-centric approach across different platforms. This future direction not only aims to broaden the scope of our findings but also to contribute to the development of more sophisticated, socially aware RecSys in the digital era.

Acknowledgments

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